rear = (rear - 1 + MAX) % MAX.

. Enqueue and Dequeue in Linear Queue.

Algorithm:

1: Start
2: If Rear == MAX-1, print "Queue Overflow" and exit.
3: If FRONT == -1, set FRONT = REAR = 0.
   Else, set REAR = REAR + 1.
4: QUEUE[REAR] = ITEM
5: End.

For Enqueue ↑

For dequeue:

1. Start
2. If FRONT==1 or FRONT>REAR, print "Underflow" and exit.
3. ITEM= QUEUE[FRONT].
4. IF FRONT=REAR, set FRONT=-1 and REAR=-1, else, set FRONT=FRONT+1.
5. End.

Circular Queue: Algorithms:

- Enqueue.

1: Start
2: If (REAR+1)%MAX = FRONT, print "Overflow" and exit.
3. If FRONT = = -1, set FRONT=REAR=0.
Else, set REAR =(REAR +1)%MAX.
4. QUEUE[REAR] = ITEM
5. END.

- Dequeue

1. Start
2. If FRONT= = -1, Print "Underflow" and exit.
3. ITEM = QUEUE[FRONT].
4. IF FRONT=REAR, set FRONT=REAR=-1
else, set FRONT=(FRONT+1)%MAX
5. End.

3. Double ended Queue :-

Algorithms :

a. Add at Beginning.

1. Start
2. If (FRONT==0 && REAR==MAX-1) or (FRONT=REAR+1) print "Overflow" and exit.
3. If FRONT==-1, FRONT=0 and REAR=0
   else if FRONT=0, set FRONT=MAX-1.
   Else set FRONT=FRONT-1.
4. DEQUE [FRONT]=ITEM.
5. END.

b. Add at End.

1. Start
2. If (FRONT==0 & REAR==MAX-1) or (FRONT=R+1), print "Overflow" and exit.
3. If FRONT==-1, set FRONT=0 and REAR=0.
   Else if, REAR==MAX-1, set REAR=0.
   Else set REAR=REAR+1.
4. DEQUE [REAR]=ITEM.
5. END.

c. Delete from Beginning.

1. Start.
2. If front==-1, print "Underflow" and exit.
3. ITEM=DEQUE [FRONT].
4. If FRONT==REAR, set FRONT=-1 ~~and~~ REAR.
   Else if FRONT=MAX-1, set FRONT=0
   else set FRONT=FRONT+1.
5. End.

d. Delete from End.

1. start.
2. If FRONT == -1, print "underflow" and exit.
3. ITEM = DEQUE[REAR].
4. If FRONT == REA, set FRONT = REAR = -1.
   else if REAR == 0, set REAR = MAX-1.
   Else, set REAR = REAR-1.
5. END.